

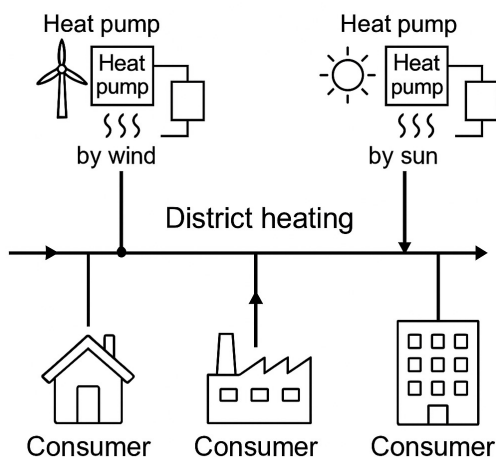
Optimal Energy-sharing and Temperature Regulation in District Heating Systems

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1. Research gap

Temperature control **challenges**
with **inaccurate** disturbances!



2. Contribution of our work

- Steady-state optimization formulation

Optimal energy sharing + Temperature deviation

= **Convex optimization problem**

- LQR framework

Optimality condition + Differential augmented dynamics = **Optimal convergence without predicting disturbance**

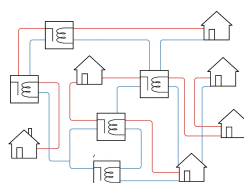
3. District heating system model

- M1: Without heat conduction

$$V\dot{T} = L_q T + P + P^{dis}$$

- M2: With heat conduction

$$V^A \dot{T} = A^A T + B^A T^{src} + W^A$$



4. Economic dispatch

Minimize Generation + Temperature deviation cost

$$\text{M1} \begin{cases} \min_{P, T} \Phi_1 = \frac{1}{2} P^T F_1 P, \\ \text{s.t. } L_q T = -(P^{dis} + P) \end{cases} \Rightarrow F^* P^* = 0$$

$$\text{M2} \begin{cases} \min_{T \in T^*} \Phi_2 = \frac{1}{2} T^T G_1 T \end{cases} \Rightarrow 1^T G_1 T^* = 0$$

$$\text{M2} \begin{cases} \min_{T^{src}, T} \Phi_3 = \frac{1}{2} (T^{srcT} F_2 T^{src} + T^{rmT} G_2 T^{rm}), \\ \text{s.t. } A^A T + W^A + B^A T^{src} = 0, T^{rm} = C T \end{cases}$$

$$F_2 T^{src} + H^T G_2 T^{rm} = 0$$

5. LQR framework and LQR-based temperature regulator

- Eliminating disturbances + Error integration:

$$\epsilon_{k+1} = A_\epsilon \epsilon_k + B_\epsilon \delta u_k, e_k = C_\epsilon \epsilon_k + D_\epsilon \delta u_k$$

$$\delta u_k = u_k - u_{k-1}, \delta T_k = T_k - T_{k-1}, \epsilon_k = \begin{bmatrix} \delta T_k \\ e_{k-1} \end{bmatrix}$$

- Transient performance:

$$\min_{\delta u_k} J = \frac{1}{2} \sum_{k=0}^{\infty} (e_k^T Q_e e_k + \delta u_k^T R_e \delta u_k)$$

Standard $\rightarrow$ LQR problem

$$\delta u_k = -K \epsilon_k$$

6. Simulation results

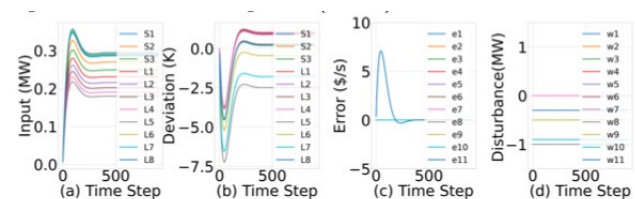
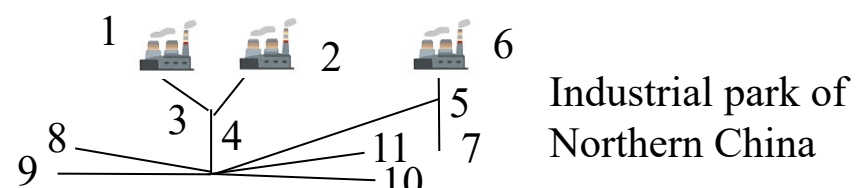


Fig. 4. (a) Controlled input, (b) Temperature deviation, (c) Optimality condition-integrated error, and (d) Load disturbances of 0MW, 0MW, 0MW, 0MW, -1MW, -0.5MW, -0.9MW, -0.3MW.

+Spike-like disturbance

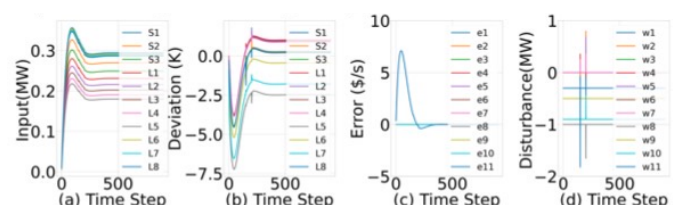


Fig. 5. (a) Control input, (b) Temperature deviation, (c) Optimality condition related error e_k , and (d) Disturbances of DHS with sudden spikes.